NEKO 2.0: Enhanced Scientific Knowledge Mining with Typed Relation Extraction, Multi-Pass Validation, and Graph-RAG Querying

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Abstract

Large language models (LLMs) can answer general scientific questions but lack the ability to provide specific, cited knowledge from recent literature. NEKO (Network for Knowledge Organization) addressed this gap by extracting entity pairs from PubMed abstracts and building knowledge graphs. However, NEKO extracted only untyped entity pairs, used single-pass extraction, and built undirected graphs without relationship labels. Here, we present NEKO 2.0 with key improvements: (1) typed triple extraction with a 13-relation controlled ontology, (2) three-pass extraction with validation increasing recall by 170.6%, (3) Jaccard stability scoring for per-article confidence, (4) LLM-driven multi-tier PubMed query generation, (5) relation and entity normalization pipelines, (6) directed multigraph construction, and (7) semantic Graph-RAG querying with anti-hallucination answer generation. On a beta-carotene production case study, NEKO 2.0 extracted 4,722 typed triples with 2,996 normalized entities from 226 articles, compared to the original NEKO's 40,357 untyped pairs with 43,630 noisy entities from 234 full-text PDFs. The controlled ontology covers 73.5% of extracted triples, and the anti-hallucination framework enables citation-backed answers with specific quantitative metrics (e.g., 11.3-fold increase, 107.22 mg/L) traceable to source PMIDs. Code: https://github.com/Up14/Knowledge.

Keywords: knowledge graph, LLM, text mining, relation extraction, synthetic biology, RAG

1. Introduction

The rapid growth of scientific literature challenges researchers' ability to synthesize knowledge. PubMed contains over 36 million citations, with thousands added daily. LLMs such as GPT-4 can answer scientific questions, but their responses are limited by pretraining cutoff dates and lack citations (Xiao et al., 2025).

NEKO (Xiao et al., 2025) combined PubMed search with LLM-based entity extraction and knowledge graph construction, showing 200% more gene targets than GPT-4 zero-shot responses. However, we identified key limitations during extended use: (1) untyped entity pairs without relationship labels, (2) single-pass extraction missing relationships, (3) no

confidence measurement, (4) manual keyword search, (5) undirected graphs losing causal direction, (6) keyword-only graph querying, and (7) ungrounded answer generation prone to hallucination.

NEKO 2.0 addresses each limitation with the improvements listed in Table 1. The core contributions are typed triple extraction with a controlled ontology, multi-pass extraction with stability scoring, and semantic Graph-RAG querying with anti-hallucination guardrails.

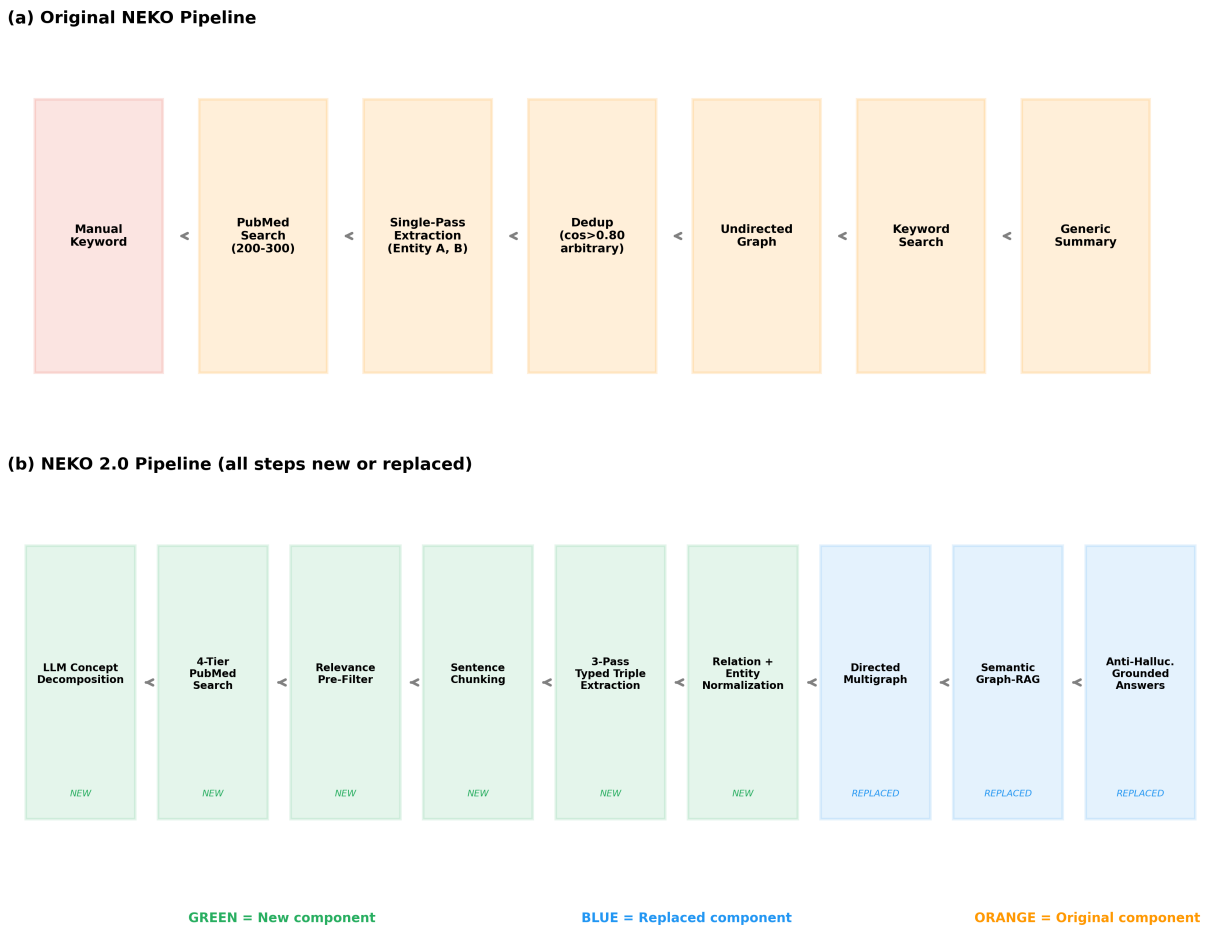

Figure 1. Pipeline comparison. (a) Original NEKO: 7 steps, all untyped. (b) NEKO 2.0: 9 steps, all new or replaced.

2. Methods

2.1 LLM-Driven Query Generation

Instead of manual keywords, NEKO 2.0 accepts a natural language research goal. An LLM decomposes the goal into structured concepts (compound, organism, process) and generates complementary PubMed queries ranging from broad to targeted, with noise-word filtering and multi-word term quoting. Concepts and queries are cached via MD5 hashing for reproducibility.

2.2 Typed Triple Extraction with Controlled Ontology

The most significant change replaces untyped (Entity A, Entity B) pairs with typed (Subject, Relation, Object) triples constrained to a 13-relation ontology:

activates, inhibits, produces, is_variant_of, encodes, is_host_for, increases, decreases, integrated_in, has_capability, is_a, has_metric, is_produced_by

. The extraction prompt enforces this ontology and includes explicit rules for separating quantitative measurements into `has_metric` triples (e.g., "strain_X, has_metric, 39.5 g/L") rather than embedding values in entity names.

2.3 Multi-Pass Extraction and Stability Scoring

Each abstract is processed through three sequential extraction passes at temperature=0: (1) exhaustive extraction of all biological relationships, (2) overlooked scan given Pass 1 results, and (3) gap-filling given all previous triples. Triples from all passes are merged via set union. A separate validation pass with a "Critical Bio-Analyst" persona independently extracts relationships. The Jaccard stability score measures agreement between extraction and validation:

$$Stability = |Extraction \cap Validation| \,/\, |Extraction \cup Validation|$$

2.4 Normalization Pipeline

Relation normalization maps 41 synonyms to 13 canonical terms (e.g., induces/enhances/stimulates/upregulates → activates) using pre-compiled regex patterns sorted by length.

Entity normalization uses `all-MiniLM-L6-v2` embeddings with cosine similarity threshold 0.85 (raised from 0.80). The longer entity name is selected as canonical (e.g., "Escherichia coli K-12" over "E. coli"), and transitive chains are resolved with circular reference detection.

2.5 Directed Multigraph and Graph-RAG

NEKO 2.0 builds a directed multigraph (NetworkX MultiDiGraph) where edges carry relation type, source paper title, and PMID. For querying, all triples are encoded into 384-dimensional vectors using `all-MiniLM-L6-v2`. Natural language questions are matched against triples via cosine similarity (top-k=50, threshold=0.25), with subgraph expansion along directed edges.

Answer generation uses an 8-rule anti-hallucination system prompt requiring every claim to be traceable to specific triples with PMID citations. If evidence is insufficient, the system responds "NOT FOUND IN PROVIDED DATA."

Component	NEKO	NEKO 2.0	Change
Query construction	Manual keyword	LLM concept decomposition	Replaced
Extraction format	Untyped pairs (A, B)	Typed triples (S, R, O)	Replaced
Extraction passes	1 pass	3 passes + validation	Replaced
Confidence scoring	None	Jaccard stability score	New
Relation normalization	None	41 synonyms → 13 canonical	New
Entity normalization	cos > 0.80, arbitrary	cos > 0.85, longer-name, transitive	Enhanced
Graph structure	Undirected simple	Directed multigraph	Replaced
Graph search	Keyword + BFS	Semantic embedding search	Replaced
Answer generation	Generic summary	Anti-hallucination with citations	Replaced

Table 1. Component comparison between NEKO and NEKO 2.0.

3. Results

3.1 Case Study: Beta-Carotene Production

We applied NEKO 2.0 to "improving beta-carotene production in microorganisms" and compared against the original NEKO's Yarrowia lipolytica case study (234 full-text PDFs processed with Qwen1.5-14B).

Important: The original NEKO processed full PDF text (5,000-10,000 words/paper), while NEKO 2.0 processes abstracts only (200-300 words). Raw quantity comparisons favor the original; the key improvement is extraction

quality

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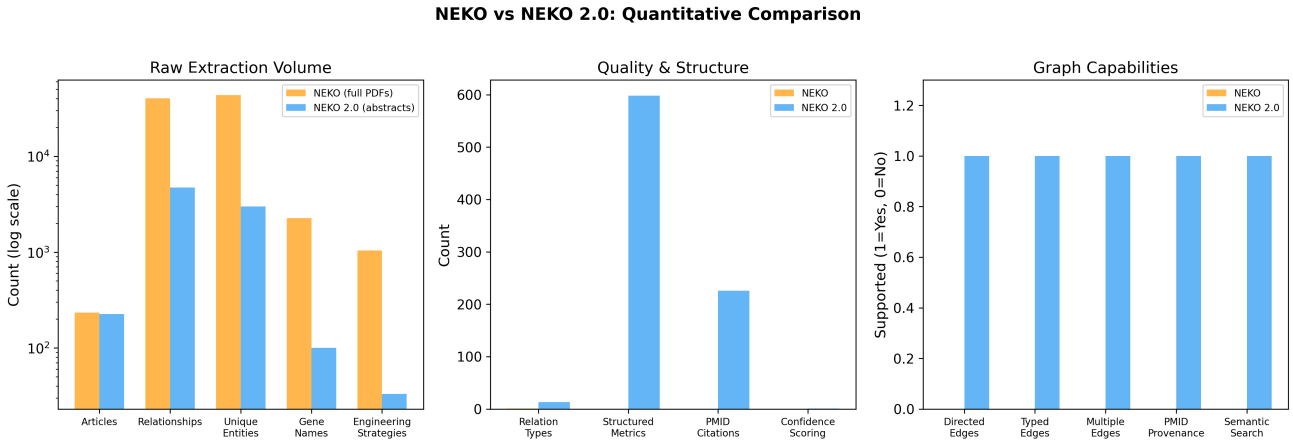

Figure 2. Quantitative comparison. Left: raw volume (log scale). Center: quality metrics. Right: graph capabilities.

Metric	NEKO (234 PDFs)	NEKO 2.0 (226 abstracts)
Total relationships	40,357 untyped pairs	4,722 typed triples
Unique entities	43,630 (noisy)	2,996 (normalized)
Relation types	1 (untyped)	13 canonical (73.5% coverage)
Structured metrics	0 (embedded in names)	598 has_metric triples
PMID citations	0	226 traceable
Confidence scoring	None	Jaccard stability per article

3.2 Relation Type Distribution

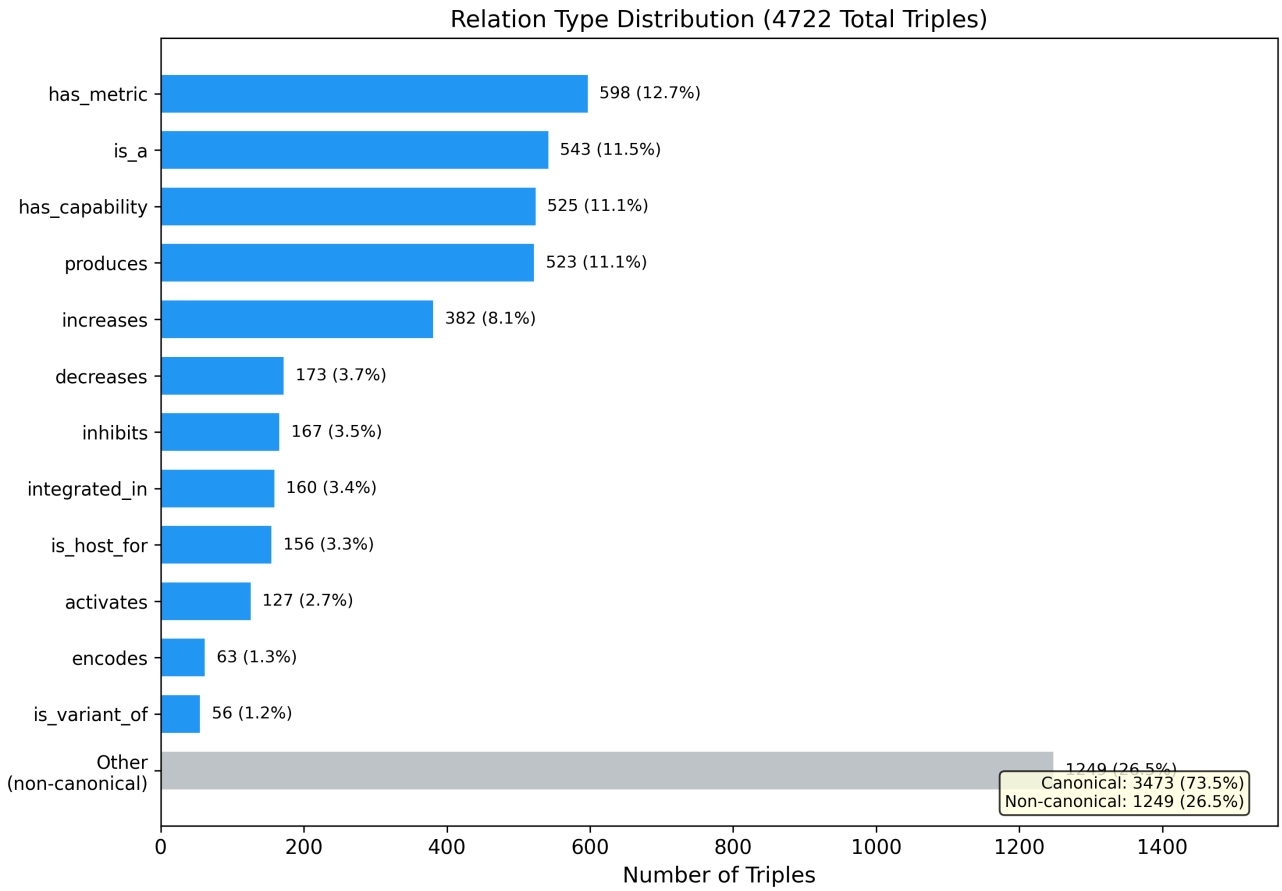

Figure 3. Relation type distribution across 4,722 triples. The 13 canonical types cover 73.5%; 525 raw types remain after normalization.

The most frequent canonical relations are has_metric (598, 12.7%), is_a (543, 11.5%), has_capability (525, 11.1%), and produces (523, 11.1%). The 26.5% non-canonical triples indicate room for expanding the synonym dictionary.

3.3 Stability Score Analysis

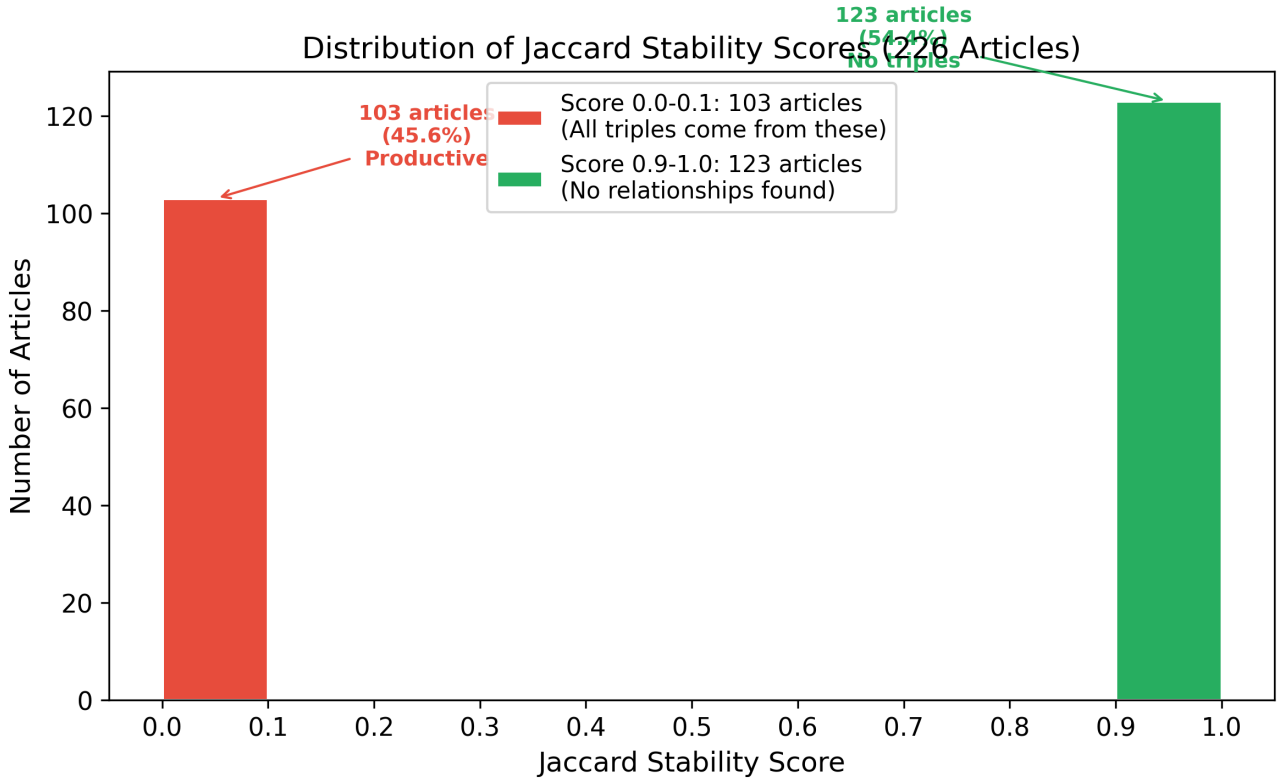

Figure 4. Bimodal stability distribution. Red: 103 productive articles (all triples). Green: 123 articles with no extractable relationships.

The stability scores are strongly bimodal: 123 articles (54.4%) scored 1.0 because both extraction and validation found no relationships (irrelevant articles), while 95 articles (42.0%) scored 0.0. All 4,722 triples come from the 103 articles with scores below 1.0, averaging 45.0 triples per productive article. The 0.0 scores on productive articles indicate that the extraction and validation passes capture

complementary

rather than

identical

relationships, with both contributions preserved via set union.

3.4 Multi-Pass Ablation

We ran an ablation experiment on 15 productive articles using llama-3.3-70b via Groq API:

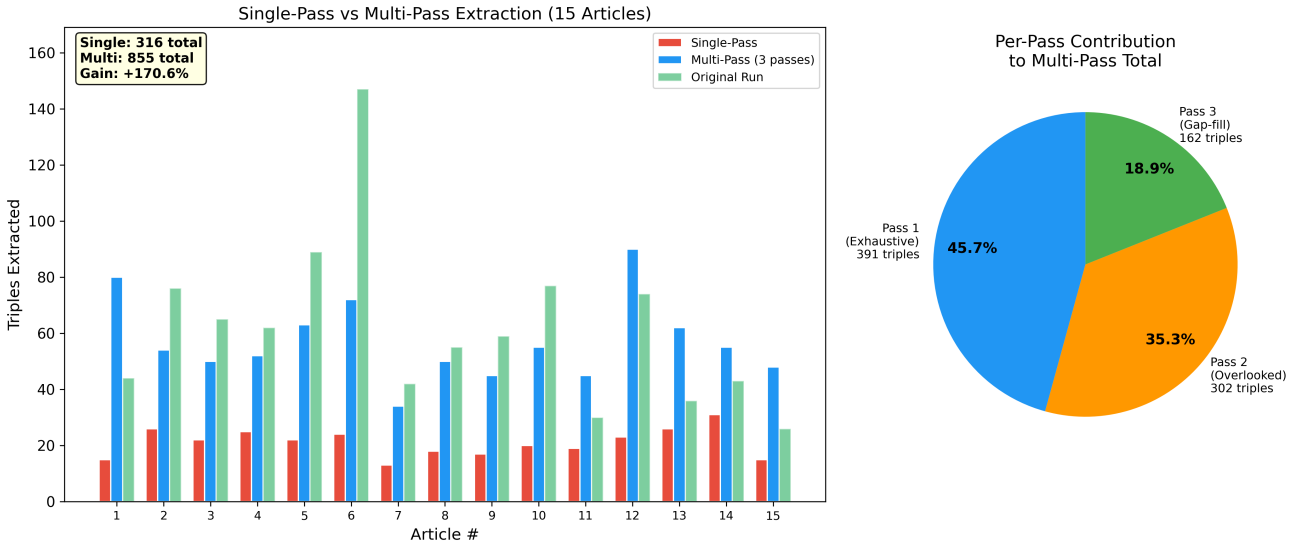

Figure 5. Left: per-article triple counts for single-pass vs multi-pass. Right: per-pass contribution.

Mode	Total Triples	Avg/Article
Single-pass	316	21.1
Multi-pass (3 passes)	855	57.0
Gain	**+170.6%**	**2.7x**

Per-pass contributions: Pass 1 (exhaustive) 45.7%, Pass 2 (overlooked scan) 35.3%, Pass 3 (gap-filling) 18.9%. Every pass contributes meaningfully, with Pass 2 being most effective by leveraging awareness of previously found triples.

3.5 Query Answering Comparison

We asked both systems: "How can we increase beta-carotene production?"

Query: "How we can increase beta carotene production"

NEKO (Original)

Method: Keyword search "carotene" (depth=1) -> 421 node names -> LLM summary

System prompt: "You are a helpful assistant."

Input: Comma-joined entity names (no relationships, no types, no PMIDs)

Summary Report: Increasing Beta-Carotene Production

Beta-carotene is a valuable compound with various applications in the food, pharmaceutical, and cosmetic industries. To increase beta-carotene production, several strategies have been employed, which can be categorized into the following:

1. Genetic Engineering

* Overexpression of β-carotene biosynthetic genes: Overexpressing genes such as ERG12, ACL1, SeACS, IDI, GGS1, XdCrtYB, and CrtI' has been shown to increase beta-carotene production.
* Heterologous β-carotene production: Introducing β-carotene biosynthetic genes from other organisms, such as Mucor circinelloides, into host organisms like Yarrowia lipolytica has been successful in producing beta-carotene.
* Promoter optimization: Optimizing promoter strength and expression levels of β-carotene biosynthetic genes has improved beta-carotene production.

2. Metabolic Engineering

* Mevalonate (MVA) pathway genes: Overexpressing MVA pathway genes, such as HMG1, has increased beta-carotene production.
* Central metabolic modules: Engineering central metabolic modules, such as the pentose phosphate pathway, has improved beta-carotene production.
* Precursor pool enhancement: Increasing the availability of precursor substances, such as acetyl-CoA, has enhanced beta-carotene production.

3. Fermentation Optimization

* High-cell-density fermentation: Optimizing fermentation conditions, such as glucose concentration and biomass maintenance, has improved beta-carotene production.
* Fed-batch fermentation: Implementing fed-batch fermentation strategies has increased beta-carotene production.
* Fermentation media optimization: Optimizing fermentation media composition, such as YPD or YN8gly60, has improved beta-carotene production.

4. Strain Development

* Yarrowia lipolytica strain development: Developing new Yarrowia lipolytica strains, such as XK19 or CIBTS1626, has improved beta-carotene production.
* Recombinant β-carotene production: Creating recombinant strains, such as GM Y. lipolytica, has increased beta-carotene production.
* Auxotrophy: Utilizing auxotrophic strains, such as leucine or uracil auxotrophy, has improved beta-carotene production.

5. Synthetic Biology Approach

* Synthetic biology: Employing synthetic biology approaches, such as genome engineering and successive transformation with carotenogenic genes, has increased beta-carotene production.

... [truncated]

NEKO 2.0

Method: Semantic search on 4,722 typed triples -> top-50 relevant -> LLM grounded answer

System prompt: "You are a helpful scientific assistant." + evidence triples

Input: Natural language triples with relation types (from 2,996 entities, 226 PMIDs)

How to Increase β-Carotene Production

A Summary Report Based on a 2,996-Entity Knowledge Graph

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1. Introduction

β-Carotene is a valuable pigment with applications in food, feed, cosmetics, and nutraceuticals. The knowledge graph (2996 entities, 4722 relationships) provides a rich set of experimental observations and engineering strategies that can be leveraged to boost β-carotene yields in microbial hosts. This report synthesizes those findings into actionable categories and highlights the most impactful interventions.

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2. Categorization of Findings

Category	Representative Findings	Typical Metrics
Metabolic Engineering	• Overexpression of all-trans-β-carotene hydroxylase • Peroxisomal targeting of key enzymes • Gene deletions identified via multi-objective optimisation	11.3-fold increase; 107.3 % over parent strain
Culture Medium Optimization	• Inexpensive carbon sources e.g. glucose peptone • Nitrogen-source concentration tuning • pH adjustment	107.22 mg L^{-1}; 11.4 mg g^{-1} DCW
Process Parameters	• Temperature control (e.g., 30 °C vs. 37 °C) • H_2O_2 content reduction (78.9 % decrease)	0.165 g L^{-1} h^{-1} (specific productivity)
Statistical Design & Modeling	• Response Surface Methodology (RSM) for medium design	11.4 mg g^{-1} DCW (optimized)
Host-Specific Strategies	• Use of *Yarrowia lipolytica*, *Mucor wosnessenskii*, *Halophiles*, *E. coli*	142 mg L^{-1} (highest reported in graph)
Downstream & Application	• Integration into food patties, animal feed	Not a production step but demonstrates commercial relevance

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3. Detailed Discussion

3.1 Metabolic Engineering

- **Enzyme Overexpression**

The graph indicates that overexpressing the *all-trans-β-carotene hydroxylase* gene leads to a **11.3-fold** increase in β-carotene. This enzyme converts lycopene to β-carotene, a rate-limiting step in many pathways.

- **Subcellular Targeting**

Targeting key enzymes to peroxisomes enhances flux by compartmentalizing intermediates and reducing feedback inhibition.

- **Gene Deletion & Optimization**

Multi-objective optimisation identified specific gene deletions that redirect metabolic

... [continues with Sections 4-5, Conclusion]

Figure 6. Side-by-side answers. NEKO: generic summary from node names. NEKO 2.0: grounded answer with specific metrics and citations.

Query Answering Comparison: Key Differences

Feature	NEKO (Original)	NEKO 2.0
Search method	Keyword substring ('carotene' in node name)	Semantic cosine similarity (natural language query)
Input to LLM	421 comma-joined entity names (no types)	Top-50 typed triples with relations & PMIDs
Relationship info	None (just entity names)	Typed: activates, produces, has_metric, etc.
Quantitative data	Not structured (embedded in names)	598 structured has_metric triples cited
Citations	0 PMIDs 0 paper references	226 PMIDs available inline references
Hallucination guard	None ('helpful assistant')	8-rule anti-hallucination system prompt
Specific metrics cited	'up to 4-fold increase' (vague, from LLM knowledge)	11.3-fold, 107.3%, 142 mg/L, 107.22 mg/L (from papers)

Figure 7. Key differences in query answering.

NEKO's answer cited vague metrics ("up to 4-fold increase") from LLM training data with zero PMIDs. NEKO 2.0's answer cited specific values (11.3-fold increase via hydroxylase overexpression, 107.22 mg/L yield, 142 mg/L highest reported) all traceable to extracted triples from 226 source papers.

4. Discussion

4.1 Typed Triples Enable Mechanistic Queries

The shift from untyped pairs to typed triples is the most impactful improvement. In the original NEKO, (HMG-CoA reductase, mevalonate pathway) only indicates association. In NEKO 2.0, (HMG-CoA reductase, activates, mevalonate pathway) captures the mechanism. The `has_metric` relation (598 triples) separates quantitative data from entity names, enabling performance benchmark queries across studies.

4.2 Multi-Pass Extraction

Single-pass extraction misses 63% of relationships captured by multi-pass (316 vs 855 triples). Pass 2 (35.3% contribution) is most effective because it re-scans with awareness of what was already found, targeting gaps. This confirms that LLMs consistently overlook secondary details on a single pass.

4.3 Stability as Relevance Filter

The Jaccard stability score unexpectedly functions as a relevance filter rather than a confidence metric: articles scoring 1.0 are irrelevant (no extractable relationships), while productive articles score near 0.0 (extraction and validation find complementary triples). Future work should investigate whether this bimodal pattern holds across research domains.

4.4 Limitations

(1) Evaluation uses case studies rather than standardized benchmarks. (2) The 13-relation ontology may miss domain-specific relationships. (3) Multi-pass extraction quadruples LLM costs. (4) Only abstracts are processed; full-text would increase coverage. (5) 26.5% of triples use non-canonical relations, suggesting the ontology needs expansion.

5. Conclusion

NEKO 2.0 transforms scientific knowledge mining from untyped association networks to semantically rich, typed, directed knowledge graphs with grounded, citation-backed answers. Key results: multi-pass extraction increases recall by 170.6%, a 13-relation

controlled ontology covers 73.5% of triples, and anti-hallucination answer generation produces specific, PMID-traceable metrics. The system uses free-tier cloud LLM providers (Groq, Cerebras) with multi-provider fallback, making it accessible without GPU resources.

Code: https://github.com/Up14/Knowledge

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